A QUASI-COMPACT QUASI-SEPARATED SCHEME WITH THE RESOLUTION PROPERTY HAS AFFINE DIAGONAL

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Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file perfect.tex; no statement or proof step was rewritten, reordered or removed. The counted span is the author environment and its own proof environment. References inside the span are handed to the assembler as navigation stubs that carry the canonical tag of the cited tag only; whether each of them is load-bearing is recorded on the stub itself.

Allowed standard prerequisites (editorial).

- The resolution property descends along approximations, so the statement reduces to Noetherian schemes. invoked at proof lines 9713-9716. Fixes the Noetherian hypothesis used later via the derived category equivalence.
- A morphism is affine as soon as the inverse images of the affine opens of the target are affine, equivalently R^1 of the pushforward of every quasi-coherent module vanishes. invoked at proof lines 9722-9730. Reduces affinity of the diagonal to the vanishing of R^1j_* for inclusions of affine opens.
- On a Noetherian scheme the derived pushforward of a quasi-coherent module is represented by a complex of quasi-coherent modules, and every quasi-coherent module is the filtered union of its coherent submodules. invoked at proof lines 9733-9748. Lets a hypothetical nonzero H^1 be detected by a coherent module F mapping nontrivially to it.
- A nonzero map to the cohomology of Rj_*G corresponds by adjunction and duality with a finite locally free module to a nonzero class in the Cech cohomology H^1 of an open, which vanishes over affines. invoked at proof lines 9750-9775. Produces the contradiction from the resolution property.

Source reference. Original proposition 01ZA: sources/stacks-chow/tags.txt, source lines 1–1. Cited at proof lines 9713-9715 for the reduction to the Noetherian case; the descent step and the whole Noetherian argument are inside the counted span.

Source reference. Original lemma 0F8B: sources/stacks-perfect/pinned-original-source.tex, source lines 9661–9661. Cited at proof lines 9714-9716 together with the approximation proposition; the Noetherian reduction is completed inside the counted span.

Source reference. Original section 01JO: sources/stacks-chow/tags.txt, source lines 1–1. Cited at proof lines 9719-9721 for the affine open cover of X x X; the reduction of affinity of the diagonal is inside the counted span.

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Source reference. Original lemma 01S8: sources/stacks-chow/tags.txt, source lines 1–1. Cited at proof lines 9722-9724 to reduce to U \cap V affine; the further reduction to R^1j_* = 0 is inside the counted span.

Source reference. Original lemma 0F83: sources/stacks-chow/tags.txt, source lines 1–1. Cited at proof lines 9726-9730 to reduce to R^1j_*G = 0; the contradiction argument against nonzero R^1 is inside the counted span.

Source reference. Original proposition 09T4: sources/stacks-perfect/pinned-original-source.tex, source lines 2026–2026. Cited at proof lines 9733-9736 to represent Rj_*G by H^bullet; the contradiction construction on the cohomology of H^bullet is inside the counted span.

Source reference. Original lemma 01PG: sources/stacks-chow/tags.txt, source lines 1–1. Cited at proof lines 9744-9748 to pick the coherent F detecting H^1; the choice of the surjection from E is inside the counted span.

Source reference. Original lemma 01XB: sources/stacks-chow/tags.txt, source lines 1–1. Cited at proof lines 9770-9773 to produce the contradiction; the adjunction and duality steps leading to the H^1 class are inside the counted span.

Source reference. Original lemma 08DQ: sources/stacks-chow/tags.txt, source lines 1–1. Cited at proof lines 9773-9775 as a reference for the dualization step; the dualization itself and the conclusion are inside the counted span.

Lemma 0.2. *Let X be a quasi-compact and quasi-separated scheme with the resolution property. Then X has affine diagonal.*

Proof. Combining Limits, Proposition 01ZA and Lemma 0F8B this reduces to the case where X is Noetherian (small detail omitted). Assume X is Noetherian. Recall that $X \times X$ is covered by the affine opens $U \times V$ for affine opens U, V of X, see Schemes, Section 01JO. Hence to show that the diagonal $\Delta : X \to X \times X$ is affine, it suffices to show that $U \cap V = \Delta^{-1}(U \times V)$ is affine for all affine opens U, V of X, see Morphisms, Lemma 01S8. In particular, it suffices to show that the inclusion morphism $j : U \to X$ is affine if U is an affine open of X. By Cohomology of Schemes, Lemma 0F83 it suffices to show that $R^1 j_* \mathcal{G} = 0$ for any quasi-coherent $\mathcal{O}_U$-module $\mathcal{G}$. By Proposition 09T4 (this is where we use that we've reduced to the Noetherian case) we can represent $Rj_*\mathcal{G}$ by a complex $\mathcal{H}^\bullet$ of quasi-coherent $\mathcal{O}_X$-modules. Assume

$$H^1(\mathcal{H}^\bullet) = \mathrm{Ker}(\mathcal{H}^1 \to \mathcal{H}^2)/\mathrm{Im}(\mathcal{H}^0 \to \mathcal{H}^1)$$

is nonzero in order to get a contradiction. Then we can find a coherent $\mathcal{O}_X$-module $\mathcal{F}$ and a map

$$\mathcal{F} \longrightarrow \mathrm{Ker}(\mathcal{H}^1 \to \mathcal{H}^2)$$

such that the composition with the projection onto $H^1(\mathcal{H}^\bullet)$ is nonzero. Namely, we can write $\mathrm{Ker}(\mathcal{H}^1 \to \mathcal{H}^2)$ as the filtered union of its coherent submodules by Properties, Lemma 01PG and then one of these will do the job. Next, we choose a finite locally free $\mathcal{O}_X$-module $\mathcal{E}$ and a surjection $\mathcal{E} \to \mathcal{F}$ using the resolution property of X. This produces a map in the derived category

$$\mathcal{E}[-1] \longrightarrow Rj_*\mathcal{G}$$

which is nonzero on cohomology sheaves and hence nonzero in $D(\mathcal{O}_X)$. By adjunction, this is the same thing as a map

$$j^*\mathcal{E}[-1] \to \mathcal{G}$$

nonzero in $D(\mathcal{O}_U)$. Since $\mathcal{E}$ is finite locally free this is the same thing as a nonzero element of

$$H^1(U, j^*\mathcal{E}^\vee \otimes_{\mathcal{O}_U} \mathcal{G})$$

where $\mathcal{E}^\vee = \mathcal{H}om_{\mathcal{O}_X}(\mathcal{E}, \mathcal{O}_X)$ is the dual finite locally free module. However, this group is zero by Cohomology of Schemes, Lemma 01XB which is the desired contradiction. (If in doubt about the step using duals, please see the more general Cohomology, Lemma 08DQ.) □